COD orders: stop the returns without killing growth

To: Founder · 30 October 2026 · Based on 1,984 past COD orders (Apr–Sep 2026). All ₹ figures are per month at ~1,650 COD orders unless stated.

The short answer

Neither “cancel risky orders” nor “ship everything” is right. Two out of three COD orders are low-risk and should ship untouched; the problem is a small, identifiable slice. I built a system that scores every COD order, then ships it, asks the customer to confirm on WhatsApp, offers a small prepaid discount, or (rarely) holds it. On September's orders it would have **cancelled 25 orders instead of the ~580 that Operations' plan cancels**, and it nets about **+₹7k/month** over six months of history (honest range: −₹11k to +₹22k, see section 4) versus **−₹100k to −₹140k** for blanket cancellation. Returns today cost us about **₹80k/month**; this system recovers roughly a third of that, not all of it — the rest cannot be separated from genuine customers without losing more than it saves.

1. What surprised me in the data

- **“First-time buyer” is not the risk — a bad address is.** Vague or placeholder addresses (“.”, “xyz”, “call me”, “near station”) return **62%** of the time (9% of orders) against **24%** for complete addresses. First-time buyers with a complete address outside the four worst states return only **20%**, and they are 53% of all first-time orders. Marketing is largely right about them.
- **Four states are the problem, not the pincode.** Bihar (54%), Uttar Pradesh (44%), Madhya Pradesh (38%) and Assam (35%) are 27% of orders but return **44%** against **23%** elsewhere (Maharashtra: 18%). “Stop COD in bad pincodes” cannot work: 1,066 distinct pincodes, none with more than 8 orders.
- **History is the strongest signal.** Repeat customers who never returned: **14%**. At least one earlier return: 45%. Two or more: **77%**.
- **Order value is U-shaped.** ₹800–1,500 orders return 21%; above ₹1,500 it jumps to **39%** (impulse buying on big baskets); under ₹800 is 27–32%.
- **What does not matter:** courier (26–31% for all five) and time of day (night 29% vs day 28%). Changing couriers will not fix this.
- **The data disagrees with the brief:** average order is **₹921**, not ₹649 (I used the real figure; profit per genuine order is ~₹480 after shipping, which makes cancelling expensive). The file also had 30 duplicate rows, 3 zero-value orders and 17 invalid phone numbers.

2. Recommended rules, and what they would have done to September

Each order gets a return-risk % from seven plain factors (address quality, state, customer history, order value, source, name and phone checks). The risk % decides the action — every cut-off is a number in config.yaml, and each decision shows its reason in plain English.

Action	Rule (default)	Sep orders*	Actually returned	Why this cut-off
Ship as-is	risk below 30%	1,104 (67%)	17%	Below ~30%, asking costs more genuine sales than the returns it stops
Confirm on WhatsApp	30%+, or shaky address/name	450 (27%)	44%	Also fixes addresses
Nudge to prepaid	clean order, risk ~39%+, value ₹500+	70 (4%)	43%	Prepaid returns only 4%; discount ₹30–75, never > half the saving
Hold / cancel	risk 75%+ (85%+ first-timers) or invalid phone	25 (2%)	60%	Cancelling pays only above ~75%

* Sample of 329 September orders scaled ×5 to 1,650. September result with real outcomes: ₹22k saved, ₹18k lost on genuine orders, ₹1k discounts = **+₹3.5k net**. The ship bucket still contains 41% of all returns: the signals are useful (hold-out AUC 0.73), not magic.

3. Operations vs Marketing, in rupees

Policy (per month)	Returns avoided	Genuine profit lost	Net	Orders cancelled
Marketing: ship everything	₹0	₹0	₹0 (₹80k return bill stays)	0
Operations: cancel risk ≥35% and block 4 states	+₹43k	−₹144k to −₹185k	**−₹101k to −₹142k**	~590 (52% were genuine)
This system (default dials)	+₹25k	−₹13k to −₹17k	**+₹7k to +₹11k**	~29

The arithmetic: one return costs ₹169 (65+65+25 + 8% damaged × ₹180). A genuine delivered order earns ~₹480 profit (₹312 at the brief's ₹649), and a first-time customer is worth another ~₹200 on average (my estimate: 25% reorder × ~2 orders × ₹400). So cancelling only pays if we are **75%+ sure** (85% for first-timers) — which is almost none of our orders. Operations is right that returns hurt (₹80k/month), but their fix loses more than it saves. Marketing is right that first-time buyers are precious: I give them a **growth guard** (never cancelled below 85% risk) and ask them to confirm instead. Both concerns are two dials in config.yaml: if you want to take Operations' side, lower hold_at_or_above_risk; for Marketing's, raise ship_below_risk — the dashboard shows the ₹ effect the next morning.

Prepaid discount: switching a risky order to prepaid saves (risk − 4%) × ₹169. At 45% risk that is ₹69, so I offer ~5% (₹30–75), capped at half the saving: ₹30 off a ₹649 order earns back ₹39 per accepted offer — about ₹1,200 per 100 offers at 30% acceptance (my guess; ₹600 at 15%).

4. What could go wrong, and how we would notice

- **Two numbers are guesses:** how many would-be returns a WhatsApp confirmation stops (I assume 55%) and how many genuine customers it loses by not replying (8%). If it stops 35% and loses 15%, the system loses ~₹11k/month; at 75% and 3% it makes ~₹22k. **Fix:** ship a random 10% of risky orders untouched for 2 weeks and compare.
- **The model drifts** (new campaigns, new customers). The dashboard compares predicted vs actual returns by risk band for orders older than 10 days and flags gaps over 12 points; re-fit monthly with scripts/calibrate.py.
- **Yesterday's ₹ are estimates** — returns appear 7–10 days later; the dashboard says so.
- **The AI part:** an LLM only reads ambiguous addresses (e.g. Hinglish landmarks). It can move the score by a small, capped amount, cannot be talked into a perfect score by text inside an address, and any failure falls back to plain rules. I could not test the live LLM call in my build environment (no network): it is tested with stand-ins; the rules-only mode is what produced every number above.
- **WhatsApp reality:** first messages must be Meta-approved templates, customers must have opted in, and Tamil/Telugu/Kannada/Bengali customers currently get English. A silent customer is cancelled at 24h (reply “yes” within a further 24h re-opens it).

5. If I had two more weeks

- Run the 10% hold-out above and replace my guesses with measured numbers (biggest lever on the decision).
- Connect the real pieces: Shopify webhook → /api/decide, a WhatsApp provider (Interakt/WATI/Gupshup), Razorpay payment links.
- Fix it at the source: house-number-required, pincode auto-complete and phone validation at checkout would remove most of the 9% “bad address” orders.
- Test a ₹30–50 COD fee in Bihar/UP instead of cancellation; add courier-by-pincode serviceability; regional-language templates.